1
Approximation with Polynomials

In many applications of mathematics, we face functions which are far more complicated than the standard functions from classical analysis. Some of these functions can not be expressed in closed form via the standard functions, and some are only known implicitly or via their graph. Think, for example, of an electric circuit, where we are measuring the current at a certain point as a function of time: the outcome might be quite complicated, and best described via a graph.

An engineer measuring the current in an electric circuit will speak about having a *signal;* for a mathematician, this just means that the output of the measurement is a function, to be called f, for which $f(x)$ equals the current at the time x. We will not give an exact definition of what is meant by a signal: for our purpose, it is sufficient to think about a signal as a manifestation of a physical event in terms of a function, or, in cases to be discussed in Chapter 5, a sequence of numbers.

Signals are usually not given directly in terms of a function; for example, they often appear via a measurement. This makes it difficult or impossible to extract exact information on the signal from the function f describing it, especially if we need to perform some calculations on f. In such cases, it is important to be able to *approximate* f with a simpler function; that is, we would like to find a function, to be called g, for which

- the relevant calculation can be performed on the function g;
- the function g is close to f, in the sense that the outcome of the calculation performed on g gives useful information about the signal described by f.

In this chapter we focus on approximation with polynomials. In its simplest version, this idea appears already in the context of finding the tangent line for a differentiable function f at a point x_0; in fact, the tangent line at x_0 is a simple function, namely, a polynomial of degree 1, which usually approximates the function f quite well in a neighborhood of x_0.

We will show how better approximations can be obtained using polynomials of higher degree. We begin with a general introduction to the idea of approximation in Section 1.1. Then we move on with approximation by polynomials in Section 1.2, which is devoted to Weierstrass' theorem: it says that *any* continuous function on a closed and bounded interval can be approximated arbitrarily well by a polynomial. In Section 1.3 we obtain a more concrete statement in Taylor's theorem, which, under certain conditions, tells us how we can choose a polynomial which approximates a given function.

1.1 Approximation of a function on an interval

Our starting point must be a precise definition of what it means that a function is approximated by another function. As we will see soon, there are in fact several different ways of defining this, and the correct definition depends on the situation at hand. Let us give a concrete example where approximation theory is needed:

Example 1.1.1 Assume that we want to calculate the integral

$$\int_0^1 e^{-x^2}dx. \tag{1.1}$$

It is well known that there does not exist a formula for the integral $\int e^{-x^2}dx$ in terms of the elementary functions; that is, we can not find (1.1) simply by integrating the function e^{-x^2} and then inserting the limits $x = 0, x = 1$. Thus, we have to find another way to estimate (1.1). This is the point where approximation theory enters the picture: in this concrete case, our program on page 1 suggests that we search for a function g for which

- $\int_0^1 g(x)dx$ can be calculated, and
- $g(x)$ is close to e^{-x^2} for $x \in [0,1]$, in the sense that we can control how much $\int_0^1 g(x)dx$ deviates from $\int_0^1 e^{-x^2}dx$.

One way of doing so is to find a positive function g which we can integrate, and for which, for some $\epsilon > 0$,

$$-\epsilon \le e^{-x^2} - g(x) \le \epsilon, \ \forall x \in [0,1],$$

or

$$-\epsilon + g(x) \leq e^{-x^2} \leq \epsilon + g(x), \ \forall x \in [0,1].$$

This, in turn, implies that

$$\int_0^1 (-\epsilon + g(x))dx \leq \int_0^1 e^{-x^2} dx \leq \int_0^1 (\epsilon + g(x))dx,$$

and therefore

$$-\epsilon + \int_0^1 g(x)dx \leq \int_0^1 e^{-x^2} dx \leq \epsilon + \int_0^1 g(x)dx.$$

Thus, $\int_0^1 g(x)dx$ gives us an approximate value for the desired integral. After developing the necessary theory in Section 1.3, we come back to the question about how to chose g in Example 1.3.8. □

The argument in Example 1.1.1 can be generalized. In fact, exactly the same consideration leads to the following:

Proposition 1.1.2 *Let $I \subset \mathbb{R}$ be a finite interval with length L. Furthermore, let f and g be integrable functions defined on I, and assume that for some $\epsilon > 0$,*

$$|f(x) - g(x)| \leq \epsilon, \ \forall x \in I. \tag{1.2}$$

Then

$$-\epsilon L + \int_I g(x)dx \leq \int_I f(x)dx \leq \epsilon L + \int_I g(x)dx.$$

Thus, if our purpose is to approximate an integral of a complicated function f, the inequality (1.2) describes a property of the function g which makes it possible to estimate the integral of f (if we can find the integral of g).

In this chapter, we consequently consider approximation in the sense of (1.2); that is, we will consider a function g to be a good approximation of f if (1.2) is satisfied with a sufficiently small value of $\epsilon > 0$.

Usually, the word *uniform approximation* is associated with the particular condition in (1.2). It is a suitable definition of approximation for many practical purposes; however, there are also cases where the fact that g approximates f in the uniform sense does not allow us to extract the desired information. For example, the condition (1.2) does not lead to any information about the difference between the derivatives f' and g':

Example 1.1.3 Let $f(x) = \cos x$ and consider for $k \in \mathbb{N}$ the functions

$$g_k(x) = \cos x + \frac{1}{k}\sin(k^2 x).$$

Then

$$|f(x) - g_k(x)| \leq \frac{1}{k}, \ \forall x \in \mathbb{R};$$

that is, by choosing k sufficiently large, we can make g_k approximate f as well as we want in the uniform sense. On the other hand, since

$$f'(x) = -\sin x, \quad g_k'(x) = -\sin x + k\cos(k^2 x),$$

we see that

$$|f'(x) - g_k'(x)| = k \ |\cos(k^2 x)|.$$

This implies that

$$\sup_{x\in\mathbb{R}} |f'(x) - g_k'(x)| = |f'(0) - g_k'(0)| = k.$$

Thus, despite the fact that g_k converges to f in the uniform sense, the distance between f' and g_k' is increasing. □

This example shows that if our purpose is to approximate the derivative of a function, a different definition of approximation is needed; uniform approximation is not suitable.

1.2 Weierstrass' theorem

We will now consider approximation in the sense of (1.2), with the extra requirement that we want g to be a polynomial. That is, given a function f defined on an interval I, we ask whether we, for a given $\epsilon > 0$, can find a polynomial P such that

$$|f(x) - P(x)| \leq \epsilon \text{ for all } x \in I.$$

A general polynomial can be written in the form

$$P(x) = a_N x^N + a_{N-1}x^{N-1} + \cdots + a_2 x^2 + a_1 x + a_0 = \sum_{n=0}^{N} a_n x^n,$$

where $a_0, \ldots, a_N$ are some numbers; if $a_N \neq 0$, the polynomial has *degree* N. It turns out that the answer to our question might be no:

Example 1.2.1 Consider the function $f :]-1,1[\to \mathbb{R}$ given by

$$f(x) = \begin{cases} 1 & \text{for } x \in]-1,0[, \\ 0 & \text{for } x \in [0,1[, \end{cases}$$

and let $\epsilon = 0.1$. Then there is no polynomial P for which

$$|f(x) - P(x)| \leq 0.1 \text{ for all } x \in]-1,1[. \tag{1.3}$$

In fact, if there was such a polynomial P, then for all $x \in]-1,0[$,

$$P(x) \geq f(x) - 0.1 \geq 0.9.$$

Since every polynomial is continuous, it follows that

$$P(0) = \lim_{x \to 0} P(x) \geq 0.9.$$

But $f(0) = 0$, so this implies that

$$|f(0) - P(0)| = |P(0)| \geq 0.9.$$

This contradicts (1.3); we conclude that no polynomial can satisfy (1.3). □

Precisely the same argument shows that we can never approximate a discontinuous function arbitrarily well by a polynomial: a single point of discontinuity is enough to make this impossible. For this reason we will only consider continuous functions in this chapter. A surprising result is that every continuous function on a closed and bounded interval can be approximated arbitrarily well with a polynomial. This is the famous *Weierstrass' theorem*:

Theorem 1.2.2 *Let $I \subset \mathbb{R}$ be a closed and bounded interval and f a continuous function defined on I. Then, for every $\epsilon > 0$ there exists a polynomial P such that*

$$|f(x) - P(x)| \leq \epsilon \text{ for all } x \in I. \tag{1.4}$$

The theorem is proved in Section A.2. The geometric meaning of the theorem is that if we surround the graph of f with a band having width 2ϵ for some $\epsilon > 0$ (see Figure 1.3.1), then there exists a polynomial that goes completely inside this band. The interesting thing about the theorem is that this is possible *regardless how little ϵ is chosen, i.e., regardless how narrow the band is.* Imagine for example how it will look like if $\epsilon = 10^{-10}$! Continuous functions can easily "behave wildly" (signals from a seismograph, medical instruments), so this is really a strong statement. See, e.g., the speech signal in Figure 1.3.2; intuitively, it is hard to imagine a polynomial which is close to such a signal over the entire interval, but Weierstrass' theorem says that such a polynomial exists.

The approximating polynomial P in (1.4) will in general depend on at least three factors, namely,

- how well we want f approximated, i.e., how small ϵ is;
- the behavior of f; strong oscillations in f usually force P to be of a high degree;
- the length of the interval I; enlarging the interval in general forces us to choose polynomials of higher degree if a certain approximation has to be obtained.

For example, it can be that for $\epsilon = 1$ we can approximate f with a second degree polynomial

$$P(x) = a_2x^2 + a_1x + a_0,$$

while for $\epsilon = 0.1$ we might need to choose a polynomial of degree 10,

$$P(x) = a_{10}x^{10} + a_9x^9 + \cdots + a_1x + a_0.$$

In principle Weierstrass' theorem makes it possible to replace complicated functions by approximating polynomials in many situations. In practice, however, it is a problem that this is only an *existence theorem:* it does not say how one can choose the polynomial that approximates a given function. The proof partly compensates for this, but the theorem is still hard to use in practice. Under extra assumptions we can reach a much more useful result, namely *Taylor's theorem.*

1.3 Taylor's theorem

In order to speak about the tangent line of a function f at a point x_0 we need to assume that the function f is differentiable at x_0. The tangent line usually approximates f well in a small neighborhood of x_0; for this reason, the tangent line is frequently called the *approximating polynomial of degree* 1. Intuitively, it is very reasonable that we should be able to obtain better approximations using polynomials of higher order. If we want to approximate f via a polynomial of second degree, it turns out to be very convenient to assume also that f' is differentiable at x_0, i.e., that f is twice differentiable at x_0. In Taylor's theorem we will approximate with polynomials of degree N for some $N \in \mathbb{N}$, and in this context we need to assume that all the derivatives up to order N exist. It turns out that there is a close link between how well we want our function f to be approximated, and the order of the needed polynomials. If we want to obtain a sequence of polynomials approximating f better and better, this means that we need to assume that f is *arbitrarily often differentiable* at the point x_0. The nth derivative of the function f at x_0 will be denoted by $f^{(n)}(x_0)$; for $n = 1, 2$ we will also use the notation $f'(x_0), f''(x_0)$.

In Taylor's theorem we will approximate the function f via polynomials P_N, $N \in \mathbb{N}$, of the form

$$\begin{aligned} &P_N(x) \\ &= f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(N)}(x_0)}{N!}(x - x_0)^N \\ &= \sum_{n=0}^{N} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n. \end{aligned}$$

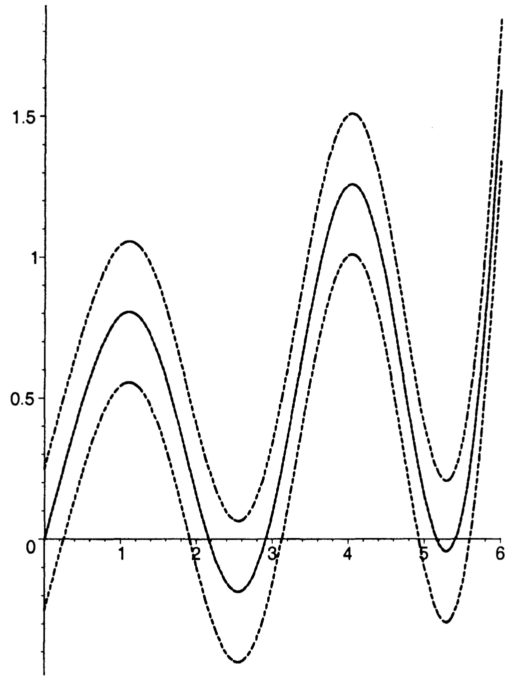

Figure 1.3.1 *A function f, (unbroken line) together with the functions $f+\epsilon, f-\epsilon$, which give an ϵ-band around the graph of f. According to Theorem 1.2.2 there exists a polynomial that goes inside this band, no matter how small ϵ is chosen, i.e., no matter how narrow the band is.*

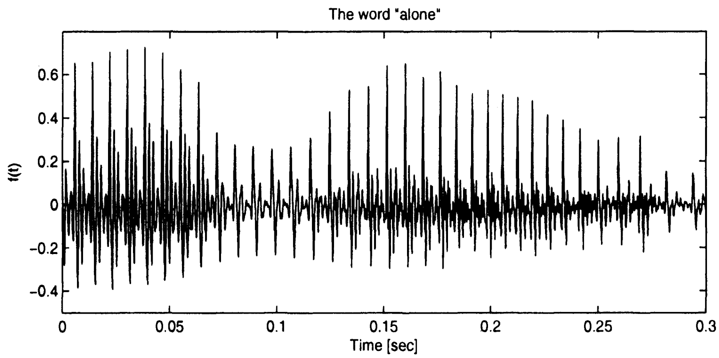

Figure 1.3.2 *A speech signal. One can think about such a signal as the current in the cable to the loudspeaker when a recording of the speech is played. The actual signal is a recording of the word "alone".*

The polynomial P_N is called the *Taylor polynomial of degree N associated to f at the point x_0* or the *Nth Taylor polynomial at x_0*. Observe that for $N = 1$,

$$P_1(x) = f(x_0) + f'(x_0)(x - x_0),$$

which is the equation for the tangent line of f at the point x_0. For higher values of N we get polynomials of higher degree, which usually approximate the function f even better for x close to x_0. Before we see a more precise version of this statement in Theorem 1.3.6 we will look at an example.

Example 1.3.3 Let us consider the function

$$f(x) = e^x, \ x \in]-1, 2[.$$

This function is arbitrarily often differentiable in $]-1, 2[$, and

$$f'(x) = f''(x) = \cdots = f^{(n)}(x) = \cdots = e^x.$$

For $N \in \mathbb{N}$ the Nth Taylor polynomial at $x_0 = 0$ is given by

$$\begin{aligned} P_N(x) &= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(N)}(0)}{N!}x^N \\ &= 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \cdots + \frac{1}{N!}x^N \\ &= \sum_{n=0}^{N} \frac{1}{n!}x^n. \end{aligned} \tag{1.5}$$

Similarly, the Nth Taylor polynomial at $x_0 = 1$ is given by

$$\begin{aligned} P_N(x) &= f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \cdots + \frac{f^{(N)}(1)}{N!}(x-1)^N \\ &= e + e(x-1) + \frac{e}{2}(x-1)^2 + \cdots + \frac{e}{N!}(x-1)^N. \end{aligned}$$

We do not unfold the terms in P_N, since this would lead to a less transparent expression. But for $N = 1$ we see that

$$P_1(x) = e + e(x-1) = ex,$$

and for $N = 2$,

$$P_2(x) = e + e(x-1) + \frac{e}{2}(x-1)^2 = \frac{e}{2}x^2 + \frac{1}{2}e.$$

The graphs for the functions f, P_1, and P_2 are shown in Figures 1.3.4 – 1.3.5. It is clear from the graphs that the second Taylor polynomial gives a better approximation of f around $x = 1$ than the first Taylor polynomial. One can reach an even better approximation by choosing a Taylor polynomial of higher degree, which however requires more calculations. In the present example it is easy to find the Taylor polynomials of higher degree; in real applications however, there will always be a tradeoff between how

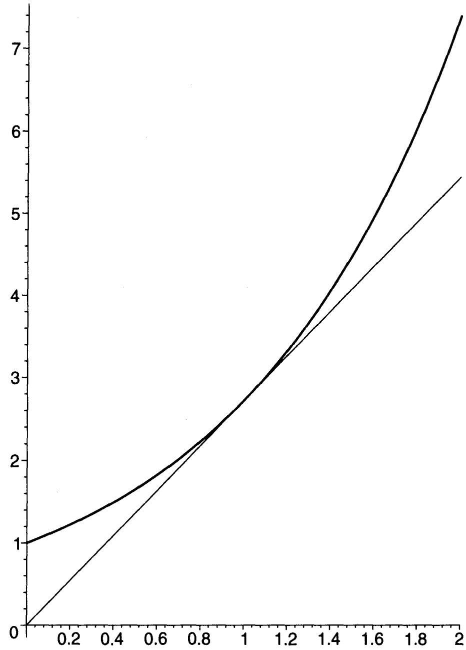

Figure 1.3.4 *The function $f(x) = e^x$ together with the first Taylor polynomial at $x_0 = 1$ (thin line).*

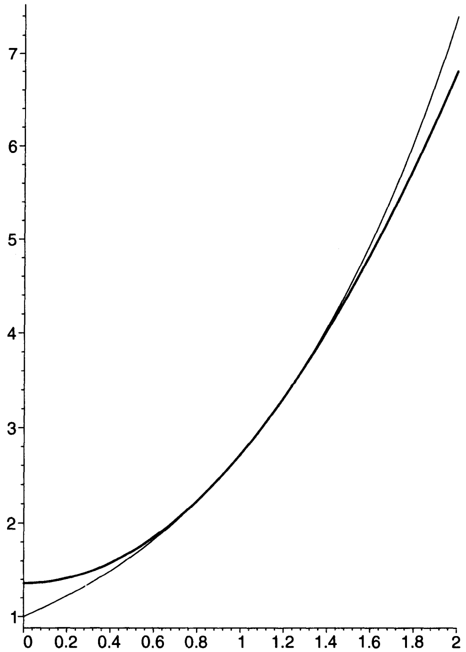

Figure 1.3.5 *The function $f(x) = e^x$ (thin curve) together with the second Taylor polynomial at $x_0 = 1$.*

good an approximation we want to obtain and how much calculation power we have at our disposal. □

We are now ready to formulate Taylor's theorem. It shows that under certain conditions, we can find a polynomial which is as close to f in a bounded interval containing x_0 as we wish:

Theorem 1.3.6 *Let $I \subset \mathbb{R}$ be an interval. Assume that the function $f : I \to \mathbb{R}$ is arbitrarily often differentiable and that there exists a constant $C > 0$ such that*

$$\left|f^{(n)}(x)\right| \leq C, \quad \textit{for all } n \in \mathbb{N} \textit{ and all } x \in I. \tag{1.6}$$

Let $x_0 \in I$. Then for all $N \in \mathbb{N}$ and all $x \in I$,

$$\left|f(x) - \sum_{n=0}^{N} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n\right| \leq \frac{C}{(N+1)!}\left|x - x_0\right|^{N+1}. \tag{1.7}$$

In particular, if I is a bounded interval, there exists for arbitrary $\epsilon > 0$ an $N_0 \in \mathbb{N}$ such that

$$\left|f(x) - \sum_{n=0}^{N} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n\right| \leq \epsilon \quad \textit{for all } x \in I \textit{ and all } N \geq N_0. \tag{1.8}$$

The proof of Theorem 1.3.6 is given in Section A.3. Comparing the result with Weierstrass' theorem, the advantage of (1.8) is that we have reached the desired approximation with a concrete polynomial. The price to pay is that the assumptions in Taylor's theorem are more restrictive.

For convenience we will usually assume that we can choose $x_0 = 0$; under the assumptions in Theorem 1.3.6 we can therefore approximate f as well as we wish with a polynomial (we suppress the dependence on N)

$$P(x) = \sum_{n=0}^{N} \frac{f^{(n)}(0)}{n!} x^n,$$

by choosing the degree N sufficiently high.

Formula (1.7) can be used to decide how many terms we should include in the Taylor polynomial in order to reach a certain approximation of a given function on a fixed interval. The following example illustrates this for the exponential function, and also shows how the number of terms depend on the considered interval and on how well we want the function to be approximated.

Example 1.3.7 We consider again the function $f(x) = e^x$. We aim at finding three values of $N \in \mathbb{N}$, namely

(i) $N \in \mathbb{N}$ such that

$$\left| e^x - \sum_{n=0}^{N} \frac{x^n}{n!} \right| \leq 0.015 \text{ for all } x \in [-1,2]. \tag{1.9}$$

(ii) $N \in \mathbb{N}$ such that

$$\left| e^x - \sum_{n=0}^{N} \frac{x^n}{n!} \right| \leq 0.0015 \text{ for all } x \in [-1,2]. \tag{1.10}$$

(iii) $N \in \mathbb{N}$ such that

$$\left| e^x - \sum_{n=0}^{N} \frac{x^n}{n!} \right| \leq 0.015 \text{ for all } x \in [-5,5]. \tag{1.11}$$

In order to do so, we first observe that the estimate (1.7) involves the constant C, which we can choose as the maximum of all derivatives of f on the considered interval. For the function f we have that $f^{(n)}(x) = e^x$, i.e., all derivatives equal the function itself.

For (i) we can thus take $C = e^2$. Now (1.7) shows that (1.9) is obtained if we choose N such that

$$\frac{e^2}{(N+1)!} 2^{N+1} \leq 0.015; \tag{1.12}$$

this is satisfied for all $N \geq 8$.

In (ii), we obtain an appropriate value for $N \in \mathbb{N}$ by replacing the number 0.015 on the right-hand side of (1.12) by 0.0015; the obtained inequality is satisfied for all $N \geq 10$.

Note than in (iii), we enlarge the interval on which we want to approximate f. In order to find an appropriate N-value in (iii), we need to replace the previous value for C with $C = e^5$; we obtain the inequality

$$\frac{e^5}{(N+1)!} 5^{N+1} \leq 0.015,$$

which is satisfied for $N \geq 19$. □

We can use the result in Example 1.3.7 to estimate the integral in Example 1.1.1:

Example 1.3.8 We return to the question of estimation of

$$\int_0^1 e^{-x^2} dx.$$

When $x \in [0,1]$, we have that $-x^2 \in [-1,0]$; thus, if we replace x by $-x^2$ in (1.9), the result in Example 1.3.7(i) tells us that

$$\left| e^{-x^2} - \sum_{n=0}^{8} \frac{(-x^2)^n}{n!} \right| \leq 0.015 \text{ for all } x \in [0,1].$$

By Proposition 1.1.2 this implies that

$$-0.015 + \int_0^1 \sum_{n=0}^{8} \frac{(-x^2)^n}{n!} dx \le \int_0^1 e^{-x^2} dx \le 0.015 + \int_0^1 \sum_{n=0}^{8} \frac{(-x^2)^n}{n!} dx.$$

Now,

$$\begin{aligned} \int_0^1 \sum_{n=0}^{8} \frac{(-x^2)^n}{n!} dx &= \left[\sum_{n=0}^{8} (-1)^n \frac{x^{2n+1}}{(2n+1)n!} \right]_0^1 \\ &= \sum_{n=0}^{8} (-1)^n \frac{1}{(2n+1)n!} \\ &= 0.747. \end{aligned} \tag{1.13}$$

Taking the small round-off error in (1.13) into consideration, this shows that

$$\int_0^1 e^{-x^2} dx = 0.747 \pm 0.016. \tag{1.14}$$

The obtained result is actually much closer to the exact value of the integral than the above tolerance indicates. For $N = 8$, the inequality (1.9) holds for all $x \in [-1, 2]$; in the arguments leading to (1.14) we only used that the inequality (1.9) holds for all $x \in [-1, 0]$, and the reader can check that this is the case already for $N = 5$ (Exercise 1.3). Using $N = 8$ rather than $N = 5$ leads to a better approximation of the exponential function, and therefore to an approximation of the integral which is better than claimed in (1.14). See Exercise 1.3 and Example 2.4.3. □

When dealing with mathematical theorems, it is essential to understand the role of the various conditions; in fact, "mathematics" can almost be defined as the science of finding the minimal condition leading to a certain conclusion. For a statement like Theorem 1.3.6, this means that we should examine the conditions once more and find out whether they are really needed. Formulated more concretely in this particular case: is it really necessary that f is infinitely often differentiable on I and that (1.6) holds, in order to reach the conclusion in (1.7)?

This discussion would take us too far at the moment, but we can indicate the answer. In order for the conclusion in (1.7) to make sense for all $n \in \mathbb{N}$, we at least need f to be arbitrarily often differentiable at x_0. Furthermore, in Section 2.4 we will see that (1.7) leads to a so-called power series representation of f in a neighborhood of x_0 (i.e., for x sufficiently close to x_0); and Theorem 2.4.10 will show us that this forces f to be arbitrarily often differentiable in that neighborhood. What remains is a discussion concerning the necessity of (1.6). For this, we refer to our later Example 2.4.12. This shows that there exists a nonconstant function $f : \mathbb{R} \to \mathbb{R}$ for which

f is arbitrarily often differentiable and $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$.

Taking $x_0 = 0$, for such a function we have

$$f(x) - \sum_{n=0}^{N} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n = f(x) - f(0) \text{ for all } \ x \in \mathbb{R}.$$

This shows that (1.7) can not hold for all $N \in \mathbb{N}$ (otherwise we would have $f(x) = f(0)$ for all $x \in \mathbb{R}$). The reason that this does not contradict Theorem 1.3.6 is that the function in Example 2.4.12 does not satisfy (1.6). This shows that the assumption (1.6) actually plays a role in the theorem; it is not just added for convenience of the proof, the conclusion might not hold if it is removed.

Example 1.3.7 shows that the better approximation we need, the higher the degree of the approximating polynomial P has to be. There is nothing that is called a "polynomial of infinite degree", but one can ask if one can obtain that $f(x) = P(x)$ if one has the right to include "infinitely many terms" in $P(x)$. So far we have not defined a sum of infinitely many numbers, but our next purpose is to show that certain functions actually can be represented in the form

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n + \cdots = \sum_{n=0}^{\infty} a_n x^n. \tag{1.15}$$

At the moment it is far from being clear how such an infinite sum should be interpreted. In the next chapter we give the formal definition, and consider functions f which can be represented this way.

1.4 Exercises

1.1 Find the first three Taylor polynomials at $x_0 = 0$ for the function

$$f(x) = e^x \cos x + \ln(x + \frac{1}{2}),$$

and plot their graphs together with the graph of f.

1.2 Find the Nth Taylor polynomial at $x_0 = 0$ for the function

$$f(x) = \ln(1 + x).$$

1.3 (i) Find $N \in \mathbb{N}$ such that

$$\left| e^x - \sum_{n=0}^{N} \frac{x^n}{n!} \right| \leq 0.015 \ \text{ for all } x \in [-1, 0].$$

(ii) Show that

$$\left| e^x - \sum_{n=0}^{8} \frac{x^n}{n!} \right| \leq 2.8 \cdot 10^{-6}, \quad \text{for all } x \in [-1, 0].$$

(iii) Show that

$$\int_0^1 e^{-x^2} dx = 0.746824 \pm 4 \cdot 10^{-6}.$$

1.4 (i) Find a polynomial $P(x) = \sum_{n=0}^{N} a_n x^n$ such that

$$\left| \sin x - \sum_{n=0}^{N} a_n x^n \right| \leq 0.1, \quad \forall x \in [0, \frac{\pi}{2}].$$

(ii) Use the result to find an approximative value of

$$\int_0^1 \sin x^4 dx,$$

with an error of at most 0.1.